

COMPUTING TECH

Mechatronics and Automated Systems

Gosford High School

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11.1 What is mechatronics?

Mechatronic devices can be described as 'smart machines'. A mechatronic engineer combines mechanical, electronic and software engineering to design and create computer-controlled machines that respond to their environment. Robots, drones and washing machines are familiar examples of mechatronic devices, however, '**mechatronics**' is set to impact our lives in far greater ways in the coming decade.

In this chapter, you will have opportunities to code, create and test your own mechatronic devices. In the projects you will build a remotely monitored irrigation system and an all-knowing automaton wizard. Although you will have plenty of experiences creating and using mechatronic devices, you will also learn about how mechatronics actually works. In the numerous Skill builders and in bonus online projects you create many interesting mechatronic devices, including a chatterbox puppet or a fascinating 'moody useless box' toy.

The importance of mechatronics

Environmental pressures, resource availability and climate change mean we need to be far more efficient in our consumption of energy and resources. Simple examples are around us already: such as escalators which only speed up when shoppers tread on a first step. Efficient lighting in many carparks and stairwells switch on only when a person enters each sector. Front doorbells with proximity sensors transmit to your phone video and audio of deliveries and visitors. Dishwashers which display how much water they are using, air-conditioners that sense when someone leaves the room, cars that monitor their maintenance and smart street lighting will become commonplace.

INFOBIT

The term 'mechatronics' was invented in 1969 by Tetsuro Mori, a senior engineer for the Japanese company, Yaskawa, by combining the words *mechanics* and *electronics*.

Mechatronics everywhere!

Look around school or home and you will see many devices which integrate mechanical, electronic and computing technologies. Whether you want to print a school assessment task, fly a drone, or blend your favourite smoothie for breakfast, mechatronics makes it possible. Modern aircraft use many mechatronic systems comprising thousands of mechanical and electrical components, each with sophisticated computer hardware and software.

INFOBIT

Hygienic handwashing: Apple Watch can automatically detect when its owner starts handwashing using its motion sensors and microphone by listening for water running and sensing relevant wrist movements. It then starts a countdown animation.

Mechatronics is making important contributions to the Internet of Things (IoT). It is a popular activity in the 'maker movement' and is the basis for many projects in STEM education. It is essential in many fields of engineering.

In this chapter as you encounter exciting current examples of mechatronic applications, use your imagination to conjure future possibilities for these technologies.



Alamy Stock Photo/OZSHOTZ

Figure 11.2 The Return and Earn scheme in New South Wales uses mechatronic reverse vending machines to reduce litter and encourage recycling.

In the industrial world, mechatronic systems are now responsible for a wide variety of tasks, from controlling assembly lines to placing frozen dinners into trays. Mechatronic systems provide the basis for modern factory automation and manufacturing where they perform a huge range of functions, including:

- welding and spray-painting
- assembly operations
- palletising and material handling
- dispensing operations
- laboratory applications
- 3D printing.

Systems thinking

Mechatronics introduces you to the components of mechatronic systems: **sensors**, '**actuators**', **microcontrollers** and the circuits that connect them. You will design and build exciting mechatronic devices in both the Skill builders and the Project.

In the following chapter you will study in more depth two specialised examples of mechatronic systems – automated '**control systems**' and robotics.

Our approach to resources

This chapter has been developed so that you can learn about mechatronics as well as learn to program the micro:bit using a general purpose language such as Python without having to purchase a great many other components.

Not all schools or students have access to expensive kits of mechatronic sensors and actuators.

The micro:bit controller already has an impressive number of sensors built in. To make this unit as accessible as possible we have designed activities using just the micro:bit and a few other sensors and actuators. You can do most activities and projects with just a micro:bit and a '**servo**'.

However, in the online resources we list some recommended micro:bit kits and provide more complex projects for those who would like to try them. Then in the next chapter we recommend some robots for that section of this focus topic.

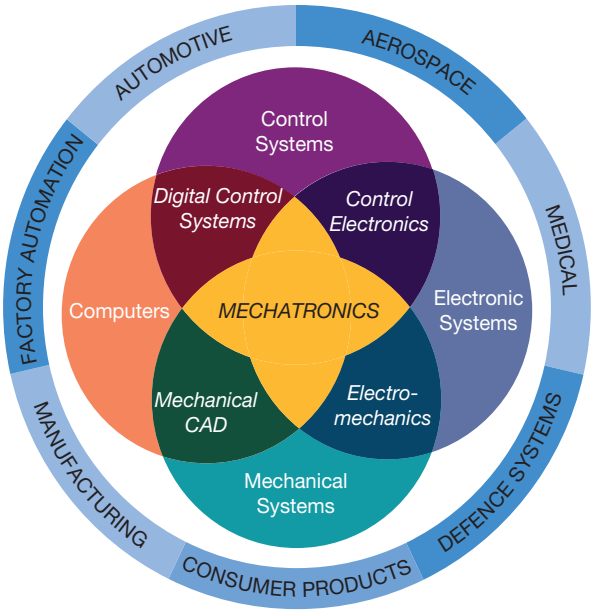


Figure 11.3 A mechatronic engineer combines mechanical, electronic and software engineering in designing and creating the wide variety of computer-controlled devices in our world and require the expertise of many specialties within these fields.

Source: Wikimedia Commons/Ahm2307/CC BY 2.5

11.2 Basic components of a mechatronic system

Sensors and actuators

The three main components which together form a mechatronic system are *sensors*, *actuators* and a *microcontroller*.

Sensors measure their environment and send this data to the device's microcontroller, manipulate it in some way depending on its programming and instruct the actuators to perform some mechanical action.

Mechatronic systems often also include feedback loops to tell the microcontroller when, say, a bottle is full or an action is complete or needs adjustment (Figure 11.4).

The parts of a mechatronic device controlled by actuators are known as *effectors*. A robot arm that welds car bodies on a production line is an example of an 'effector' and the motors that move it are its actuators.

Microcontrollers

Microcontrollers are the brains of mechatronic systems. They are small dedicated computers on a single board. They receive and process data and interact with a range of system components including **accelerometers**, **infrared** sensors, motors, pumps and valves.

Although they are computers, microcontrollers are slow and have very little memory. However, they are cheap and very easy to interface with real world devices. You will find them in most electronics these days, from microwave ovens to televisions.

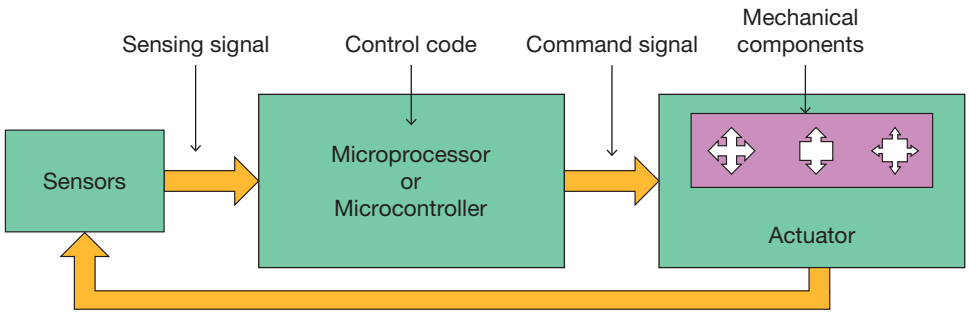
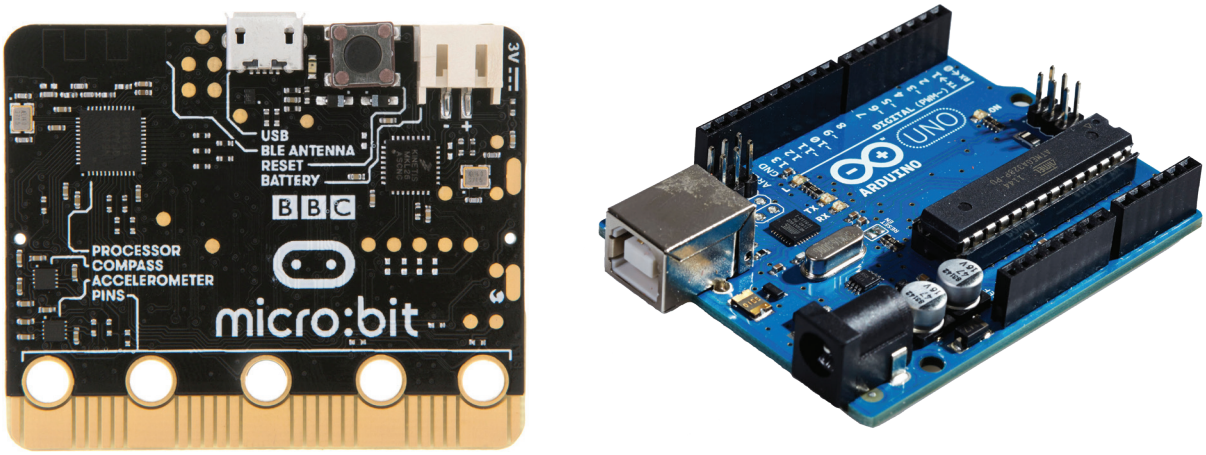


Figure 11.4 Mechatronic systems use sensors to send data to microprocessors which contain programs which instruct actuators.



Alamy Stock Photo/urbanbuzz

Alamy Stock Photo/Premier

Figure 11.5 Popular microcontrollers: a micro:bit (left) and Arduino (right). Microcontrollers can be described as miniature computer motherboards and are integral components in contemporary mechatronic systems.

ACTIVITIES

Identify

1. What two words are combined to form the word mechatronics?
2. Identify and describe an application of mechatronics found in most homes.
3. What is the function of a microcontroller in mechatronic devices?
4. Name a microcontroller used in prototyping mechatronic devices.

Analyse

5. Outline the difference between an effector and an actuator and provide an example of each.
6. Is there a difference between robotics and mechatronics? Explain your answer.

Extend

7. Research the operation of a typical desktop 3D printer, draw a diagram, identify its sensors and actuators and describe their functions.

ACTIVITIES: WHAT IS MECHATRONICS?

1. Distinguish between an effector and an actuator.

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2. Describe the function of a microcontroller in mechatronic devices. Provide an example to support your response.

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3. Explain the difference between robotics and mechatronics.

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11.3 Sensors

As human beings our senses are an essential part of our lives. They provide us with the ability to interpret our world. Our eyes give us sight, our ears interpret sound, our nose senses smell, our skin responds to touch and temperature and our mouth experiences taste. Although mechatronic sensors are modelled on nature, they typically cannot match the amazing range and precision of our human senses.

Sensors are input devices. Mechatronic devices use them to collect real world data such as temperature, light, noise, acceleration and movement. Sensors send this data to a central processing unit ('**microprocessor**') as input for programming functions which produce an appropriate response, perhaps by moving an output device, such as an actuator, or switching something on.

Types of sensors

Sensors obtain information from their environment. The type of sensor used will depend on the type of data required by the system. Sensors can be analogue or digital and may be classified as either active or passive.

Active sensors require physical interaction between the sensor and another device. These sensors use sound waves rather than light or visual data. For example, active sensors may be used to track people in a building or they may be used to trace the location of ships at sea.

Passive sensors require a change in the environment in order to operate. Most security systems use sensors called passive infrared detectors (PIRs) to detect intruders and turn on an alarm. PIRs are electronic devices that detect body heat when intruders enter or move around a protected area.

Table 11.1 Various forms of sensors are used to measure different types of data.

Type of sensor	Function
Temperature settings	Measure current temperature and may include control units, such as those used to control temperature in air conditioning units
Motion and pressure sensors	Detect sudden changes of movement
Optical/light sensors	Detect changes in light or changes in colour
Chemical sensors	Detect the presence and quantity of various chemicals

INFOBIT

Touchless thermometers are able to detect body temperature from a distance, of critical importance in detecting potential COVID infection. A person stands in front of a small lens that focuses the infra red radiation from our body and the temperature sensor displays the reading within seconds.

Common applications of sensors

Security alarms

There are many types of security alarms used in homes and commercial buildings. You may have one in your home. A common design is an alarm that uses a detector to pick up unexpected movements in its path. These types of systems have two main components:

1 a source of light, such as a laser beam
2 a photoelectric sensor.

The system may use reflection where the beam bounces back unless interrupted, or an uninterrupted beam directed at a receiving sensor.

In a home security system, a light beam (visible or infrared) may be aimed across a passageway. When somebody walks between the light source and the sensor, the path of the beam is momentarily blocked. The sensor registers a drop in light levels and sends a signal to the control box. This activates a timer and allows the homeowner to enter a security code. If the security code is not entered, an alarm sounds.



iStock.com/vchal

Figure 11.6 Passive infrared motion detectors (PIR) are often connected to security alarms.

Automatic doors

Automatic doors use a sensor to open doors at the required time. The sensor is often placed at the head of the door. It is passive and may be an '**ultrasonic**' device or an infrared detector. When the presence of a person is detected, a message is sent to the controller to open the door.



Shutterstock.com/Hatsaniuk

Figure 11.7 Automatic doors use optical or motion detection sensors to activate their motorised opening and closing mechanisms.

Vehicles

Late-model cars contain an astonishing array of sensors to allow engines and electronic systems to run at peak efficiency. Sensors are used to measure speed, revs, lane and obstacle proximity, fuel level, rain, tyre pressure, engine temperature, battery condition and to respond to collisions by inflating airbags.



iStock.com/teddyleung

Figure 11.8 Engine control units accept inputs from a range of sensors and display a variety of information about a vehicle's performance.

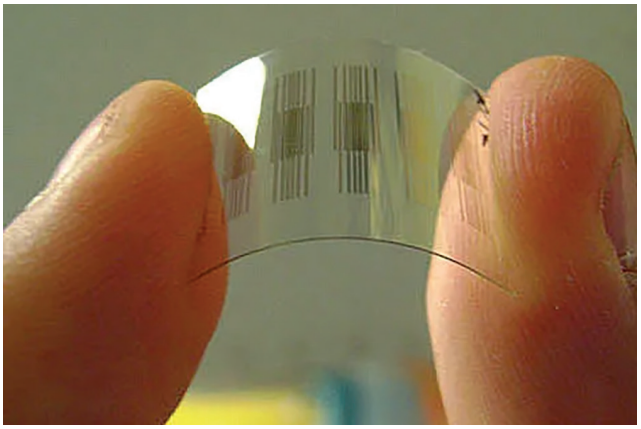
INFOBIT

Developments in chemical sensors now enable doctors to perform a blood test and obtain immediate results. Hand-held sensors allow physicians to get an immediate on-site chemical analysis, without having to wait for laboratory reports.

Nanosensors

Tiny sensors called 'nanosensors' offer great potential, particularly in medicine.

Development of these nano-enabled sensors is in the early stages, but the technology is expected to be a strong force in the treatment of a range of illnesses from the common cold to cancer. The present era of nanotechnology has reached a stage where scientists are able to develop tiny programmable machines that can work inside a patient's body.



Reprinted from Nanotechnology and Functional Materials for Engineers, Yaser Dahman, Chapter 6 - Nanopolymers, Copyright 2017, with permission from Elsevier.

Figure 11.9 A nanosensor embedded on a piece of plastic. Sensors such as these can detect explosives and toxins in water.

Nanosensors can measure chemicals and biological agents present in the air and were used during the Beijing Olympics to measure air pollution. One great strength of nanosensors is they can detect incredibly small amounts of viruses, bacteria and fungi. Cantilever (bending) nanosensors can even detect changes at the genetic level as shown in Figure 11.10.

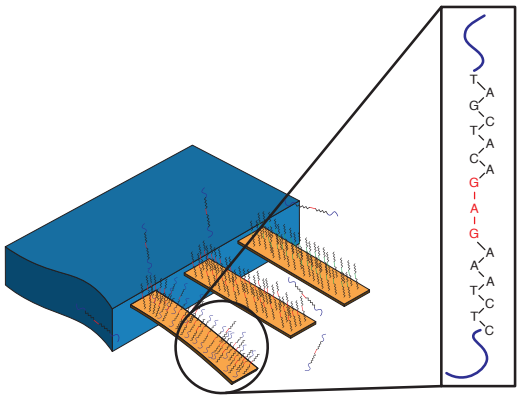


Figure 11.10 In this nanosensor used for DNA testing, the cantilever 'arm' on the left is encoded with a particular sequence for a mutation. If this same sequence is present in a sample being tested, the corresponding segment of RNA binds to it, causing it to bend and change its conductivity. This provides clear evidence that the genetic mutation is present.

Source: University of Basel, Department of Physics

9780170442787

In the future, nanotechnology may enable engineers to construct sophisticated nanorobots that can navigate the human body, transport important molecules, manipulate microscopic objects and communicate with physicians by way of miniature sensors, motors, manipulators, power generators and molecular-scale computers.

KNOWLEDGE BUILDER

Exploring your phone's sensors


Systems thinking

Although not a mechatronic device, your phone is an excellent way to familiarise yourself with the sensors used by mechatronic devices.

Table 11.2 lists some of the input sensors in a computer most of us use every day – a mobile phone. Phones have more sensors than a desktop computer! Mobile phones are not mechatronic devices but they are a great way to become familiar with various sensors we need to use later.

Individually, or as part of a group, copy and complete this table. If in a group, use a collaborative tool such as Google sheets to record responses.

Table 11.2 Some of the input sensors in a mobile phone.

Sensor	What does it measure?	What is it used for?
Accelerometer		
Ambient light		
Front camera		
Rear camera		
Proximity		
Gyroscope		
Magnetometer		
Barometer		
NFC		
Touch fingerprint		
Temperature		
Moisture		
Face ID		

SKILL BUILDER

Phone and tablet sensors in action

There are exciting and easy-to-use free apps for Apple and Android phones and tablets such as iPads which access and display data from the phone's many sensors.

1 Download *Arduino Science Journal* (Figure 11.11) (available for both Apple and Arduino). Press the plus symbol to create an untitled experiment. Press the pencil edit icon then select Sensors. Add multiple sensors to examine data for acceleration in each of x, y and z directions, barometric pressure, brightness, compass ('**magnetometer**' + 'accelerometer'), linear acceleration (which ignores gravity), magnetometer and sound intensity. Click the black 'i' symbol to read background explanations for each of these.

2 Make a data recording using *Arduino Science Journal* by clicking the Record button. After a few seconds moving your phone around press Stop. Click the back arrow at the top to see the actual recording. Select it. Click 'Review' at the top to see the actual recording. Click the three dots and export the data by selecting Save to files (the file will be in .csv format for spreadsheet analysis). It is also possible to send the whole experiment using Archive. This will be a file that can be imported by any device that has Science Journal installed.

3 Download the app *Sensors Toolbox* (Figure 11.11). Examine the accelerometer, 'gyroscope' and magnetometer data and graphs. Use the column graph icons to view data in the form of graphs and use the top right drop-down icon to export it in CSV format by clicking 'Share'.

4 Compare the UI and UX of these two apps. See Chapter 7.

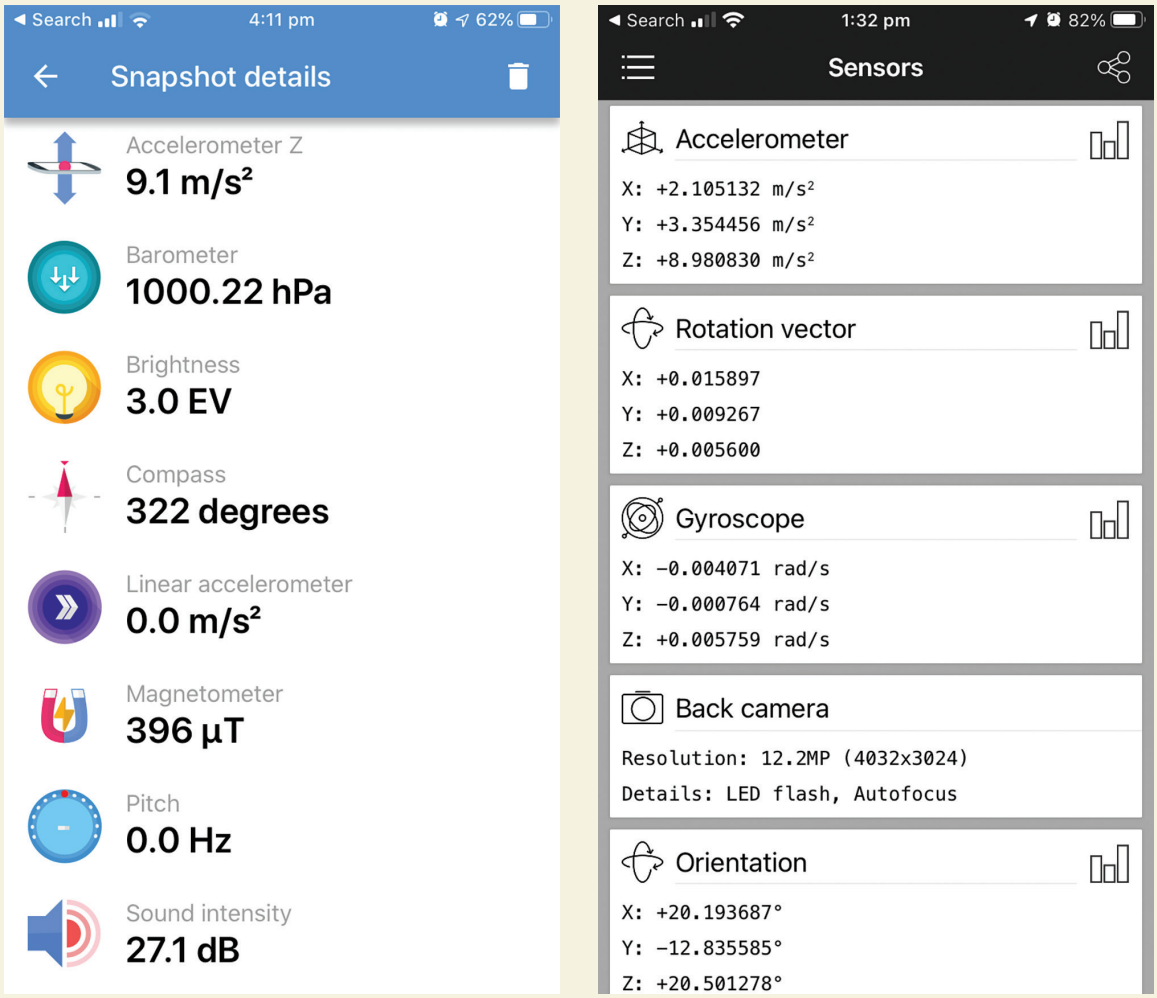


Figure 11.11 Screenshots showing the apps *Arduino Science Journal* and *Sensors Toolbox*.

ACTIVITIES

Identify

1 Identify five sensors found in a typical mobile phone.

2 What is the function of the magnetometer sensor in a phone?

3 What is one advantage of nanosensors over regular sensors?

Analyse

4 Describe the difference between an active and a passive sensor and give an example of each.

5 Describe the mechanism of airbag operation in a car accident, identifying the sensors and actuators involved.

Extend

6 Research nanosensors and describe one medical application for them.

11.4 Accelerometers

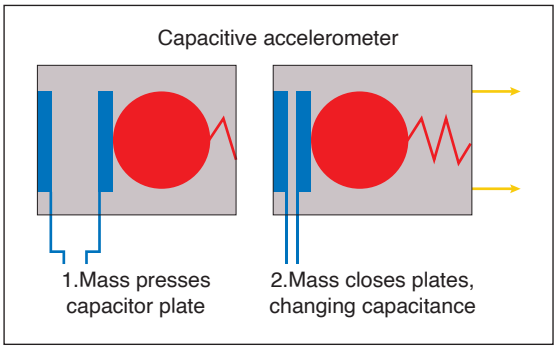
One sensor is used so widely we will examine it here in more detail. This is the accelerometer. Accelerometers are used to measure acceleration typically as three separate measurements: along each of the x, y and z axes.

Accelerometers have a great many applications in mechatronics:

- trigger safety airbags in car accidents
- in combination with **gyroscope** sensors, they keep robots upright and drones horizontal
- undo typing errors on phones by recognising the shake gesture
- count steps
- detect earthquakes
- measure acceleration during rocket lift-offs
- Nike running shoes use them to calculate the amount of time the shoe is on the ground versus the time it is in the air and mobile devices then convert the data into an accurate measure of the runner's speed
- in video game controllers, accelerometers sense the angle a controller is held when simulating swordplay
- mobile phones use accelerometers more than any other sensor – among many functions performed, they recognise a phone's tilt, adjust its display from portrait to landscape and eliminate unsteadiness and shake when taking photos.

How do accelerometers work?

Accelerometers are examples of miniature components called microelectromechanical systems (MEMS). Common accelerometers work by using a weight on a spring which presses against capacitive plates (Figure 11.12). Capacitors are electronic components which are able to store electrical charge for a short time. As two plates move nearer each other, the capacitance between them changes. From these changes, acceleration can be measured.



But these plates actually look more like this:

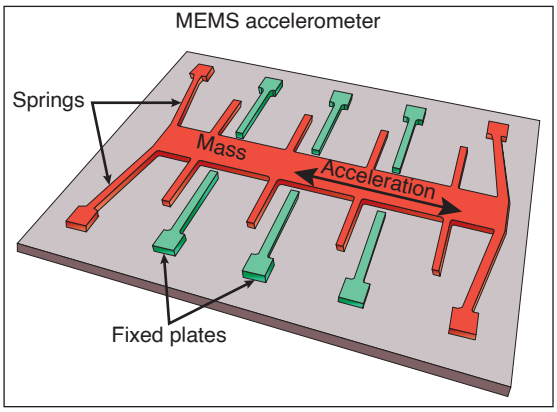


Figure 11.12 An accelerometer senses acceleration by measuring changes in the capacitance of its charged plates as they move.

KNOWLEDGE BUILDER

Understanding acceleration sensors

In the next section, you will explore accelerometers using a micro:bit microcontroller board – you can do this even if you do not own one as a simulated one is provided on the website! First you need to understand how acceleration is measured.

The tiny accelerometer on this micro:bit can be seen on its reverse side, see Figure 11.13.

From physics, you know that acceleration is produced when an object is acted on by an unbalanced force. In an accelerometer, these forces move tiny accelerometer weights in the x, y and z directions.

In Figure 11.13:

- x represents acceleration left and right along the x axis
- y represents acceleration forward and backward along the y axis
- z represents acceleration up and down along the z axis.

CASE STUDY

When sensors go wrong

In 2019, following a second catastrophic crash in five months of a Boeing 737 Max, this time belonging to Ethiopian Airlines, aviation authorities grounded the entire global fleet of 737 Max aircraft. A preliminary report stated the pilots were unable to prevent the plane repeatedly nosediving. The captain and first officer followed safety procedures recommended by Boeing but couldn't stop their aircraft going into a fatal dive shortly after take-off. All 157 passengers were killed.

A stall can happen when an aeroplane flies at too steep an angle (Figure 11.22). This reduces the lift generated by the wings, potentially making the plane drop. To recover from a stall, a pilot would normally push the aeroplane's nose down but in the 737 Max, an automated system was designed to lift it back to 'normal' flight position and repeat the process if it detects the plane is still tilted at too great an angle.

The report stated that a short time after take-off, two sensors that measure the tilt angle of the aeroplane's flight began to record different readings (see Figure 11.23).

If two sensors gave conflicting data, the system's algorithms were programmed to select the worst scenario. The system interpreted this as a dangerously steep climb and repeatedly pushed the nose of the aeroplane down.

Systems thinking

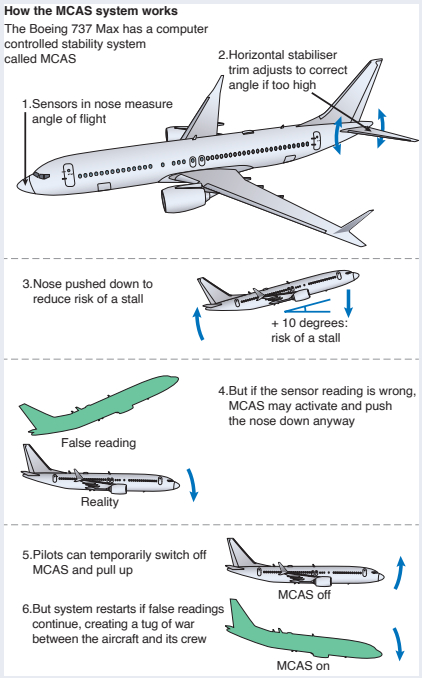


Figure 11.22 How sensors in the MCAS stability system work.
Source: Boeing, The Air Current and BBC

The report suggested the pilots followed correct steps to disengage the system, but even after manually steadying the aeroplane, the system was designed to reset itself and pushed the nose down until the aircraft eventually crashed.

Adapted from 'Boeing 737 Max: What went wrong?', BBC News, 5 April 2019.

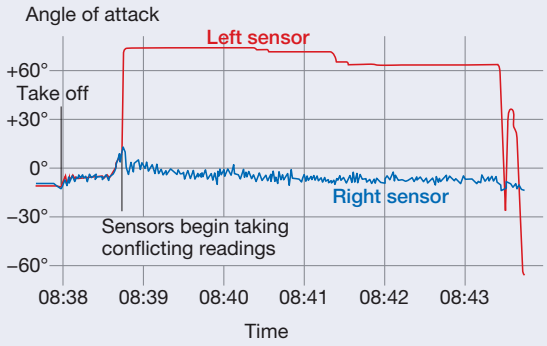


Figure 11.23 Different readings register shortly after the aeroplane's take-off.
Source: Ethiopian Aircraft Accident Investigation Bureau and BBC

You will now simulate this accident in the following Skill builder!

Questions

1. Explain the preliminary investigation's findings in your own words.
2. Who or what would you say was the cause of this tragedy?
3. Assuming the initial findings are correct, identify three safeguards that might prevent such an accident.
4. It would be possible to disguise the cause of the accident by altering data collected during the flight. Although not an issue in this case, what precautions could be taken to guard data security in such a critical control system?
5. Similar challenges exist in building a completely autonomous (self-driving) car. These may involve mechanical, electronic or computer programming issues. The greatest challenge according to the researchers at the University of Sydney is anticipating unpredictable human behaviour. Identify five scenarios that might result in an accident and suggest a suitable response algorithm for each.

SKILL BUILDER

ACTIVITIES

Identify

1 Identify three different applications of accelerometers in mechatronic devices.
2 What common electronic component is used by an accelerometer?
3 Sketch the three axes for an accelerometer and indicate the gravitational force.
4 'RADAR' and 'LiDAR' are both used as sensors in autonomous vehicles. What do these acronyms stand for?

Analyse

5 Describe how an accelerometer sensor works.
6 Describe how x, y and z measures of acceleration taken by an accelerometer are combined to create one overall measure.

Extend

7 Describe how pitch, roll and yaw can be calculated using an accelerometer.

11.5 Data logging

Data loggers are electronic devices using built-in sensors to record data over time. These mechatronic devices can either store the data internally or transmit it to another device, usually a personal computer. By graphing the data, trends can be observed over time that would otherwise not be observed.

Data logging is used in many applications. Examples include weather stations, road traffic, wildlife counts, soil moisture, dam water levels, solar panel electricity production and pollution measurements.

In the following Skill builder, you will use a micro:bit as a data logger to recreate the previous case study.

11.6 Distance sensors

Another important category of mechatronics sensor measures distance.

Your phone has a proximity sensor near the top of the screen which turns off the display when you are holding it near your ear to eliminate the annoyance of a bright screen light and your cheek touching onscreen buttons. This operates only when the phone's accelerometer detects the phone is upright (another use for the accelerometer!) otherwise it would trigger constantly. Your phone's proximity sensor emits a beam of infrared light and uses the reflected light to register if an object is within range.

Proximity sensors are common on production lines where they are used to sense when a manufactured object is aligned at a certain position for performing some operation (such as filling a bottle with soft drink or placing components in a device). These work by detecting changes in capacitance between the sensor and an object.

There are four main types of distance sensors (Figure 11.33):

Infrared sensors measure the distance to objects. As the light cannot be seen by humans, these can be used discreetly around our homes to trigger alarms or turn on lights.

Ultrasonic sensors measure distance using sound waves. They send out a sound wave at a known frequency and measure time taken for the wave to bounce back, converting this to a distance.

RADAR (RAdio Detection And Ranging) sensors use radio waves to evaluate distance or velocity by measuring the time for the reflected wave to bounce back.

LiDAR (Light Detection And Ranging) sensors illuminate a target with laser light and measure the reflected light to evaluate distance. Algorithms can use this to build 3D representations of distant objects.

ACTIVITIES: SENSORS

1. Describe ONE use of an accelerometer. Use an example to support your response.

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2. Explain how a data logger could support weather predictions.

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11.7 Actuators

What is an actuator?

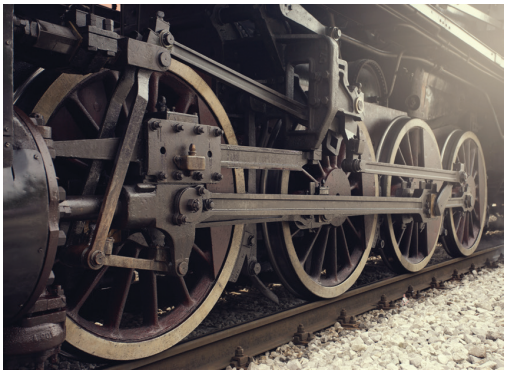
Actuators are mechanical devices used to move things. In mechatronic devices, they act as output devices.

In simple terms, actuators represent the muscles of automated systems. They convert commands into actions and perform the physical work of a control system.

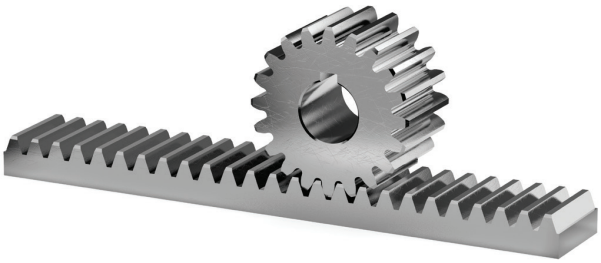
Actuators are not always electrically driven. They may also be hydraulic (operated by liquid pressure) or pneumatic (operated by air pressure) and most are driven by software. An actuator connects a control system to its environment.

Many actions that were mechanical in the past have been replaced by electronic actuators. Some examples are automatic car window winders, windscreen washers, automatic boot opening and the operation of closing and opening valves, such as in automatic washing machines.

The most common action that actuators perform is rotation, although this can be changed into linear motion using a slider and a crank. In the case of steam locomotives, the action is performed in reverse, or rack and pinion (see Figure 11.37).



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Shutterstock.com/AndreyPreakt

Figure 11.37 Steam engine wheels turn linear into rotational movement using a crank and rack and pinion gears perform the reverse function.

Weblink
How Slider Crank Mechanism Works

INFOBIT

Researchers strive to imitate characteristics of the actuators our bodies use – our muscles. Soft actuators are being developed to harvest fruit or manipulating organs when operating on the body. This will open up a wide range of new applications for mechatronics. Soft actuators move flexibly through changes at a microscopic level. Shape memory polymer (SMP) actuators are similar to our muscles, providing a response to a range of stimuli such as light, electrical, magnetic, heat, pH and moisture changes.

An SMP can be used as a gripper actuator.

Motors

Motors are easily the most common actuator. There are three very different types of motors used widely in mechatronics: DC motors, servos and '**stepper motors**'.

Weblink
Gripper based on dielectric minimum energy structures

DC motors

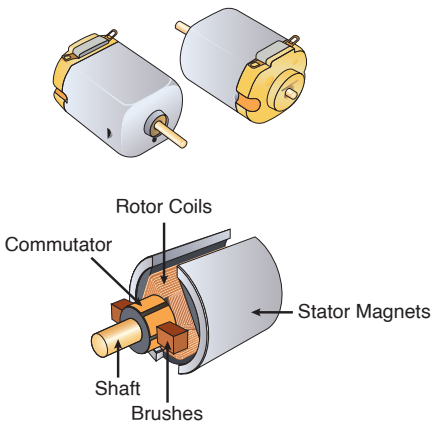


Figure 11.38 DC motors provide constant rotation.

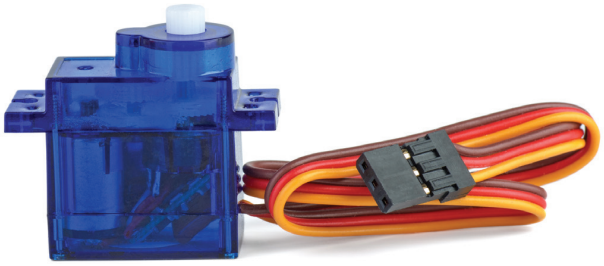
Weblink
How DC Motors Work

Most battery powered moving toys use DC motors. DC motors are constructed of many windings of copper wire around a rotor. When this rotor experiences the opposing magnetic field of the two permanent magnets around it, it is forced to rotate.

This also works in reverse! If a DC motor of this type is spun in reverse (such as by wind by attached fan blades) it produces a small current. (We will use this characteristic in a later Skill builder.)

This animation provides a detailed explanation of how DC motors work.

Servos



iStock.com/digitalr

Figure 11.39 A popular hobby servo motor used for prototype mechatronics systems.

Hobby servo motors like the ones you will be using in many of your projects are different from standard DC electric motors as a voltage controls the *position* of the motors axis rather than the *speed* of rotation. A servo motor does not rotate constantly in the way a DC motor does (although it does have a DC motor inside it). Instead, it rotates to a set angle and holds that angle.

Weblink
How does a Stepper Motor work?

This behaviour is needed in a wide range of mechatronic applications. Examples include focusing for cameras, window and door openers, robotic arms, steering for radio controlled vehicles, drones and keeping cameras level.

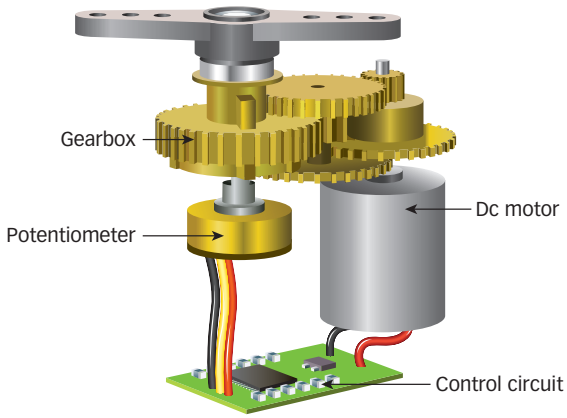


Figure 11.40 A servo motor holds its rotational position by using feedback from a potentiometer.

To achieve this behaviour, a servo motor has a 'closed-loop' control system that uses feedback to control its final position.

One feedback technique uses a potentiometer (a variable resistor) attached to the output axle of a regular DC motor. As the axle turns it changes the voltage delivered to a control circuit by turning this potentiometer. This voltage will be proportional to the rotation of the large gear and can be measured against the voltage originally sent to the servo, thus causing the motor to rotate until the two are the same. This explains why hobby servos sometimes 'tremble' or 'stutter' if they do not have sufficient voltage and why their motion is typically limited to 180 degrees (some servos are capable of 360-degree rotation).

You will be using servos soon in a Skill builder and in the Project at the end of this chapter.

Stepper motors

Stepper motors also provide precise movement to a specified location but without the need for a feedback loop. The rotational movement can be controlled to an accurate number of degrees. Stepper motors are useful in automated systems when very precise movements are required (as in 3D printing).

The motor's shaft can be commanded to move and hold any one of its many step positions without requiring feedback as does the servo. The operation is simple when seen, but difficult to explain in words. The best way to understand how it does this is by viewing an animation.

Stepper motors are used in 3D printers and other devices that need to move in a precise and controlled manner. Servo motors are used in devices that need to move smoothly.

Table 11.3 An overview of common actuators.

Type of actuator	Function
Solenoids	Used to move an arm through a small movement
Electric motors	Provide smooth, continuous rotational motion in a clockwise and anticlockwise direction
Servos	Provide continuous angular movement to a specified location, often from 0 to 180 degrees using closed loop control and high torque at low speeds. Lower power needed
Stepper motors	Provide precise stepped angular movement without feedback control and high torque at high speeds. Higher power needed
Relays	Act as a switch to control mechanical movement
Pumps	Used to force liquid or gas into piston chambers to cause mechanical movement



Shutterstock.com/monkey business images

Figure 11.41 Stepper motors control the precise movements required by 3D printers.

Solenoids

A '**solenoid**' consists of a magnetised iron rod surrounded by a tightly bound coil of wire. When an electric current passes through the wire, a strong magnetic force is produced that is strong enough to move the rod. The principle is the same as that used in a motor. A push movement may be used to connect the starter motor of a car. A pull movement may be used to open the valve of an automatic watering system.

Relays

Relays are sometimes called 'switches'. They consist of a solenoid connected to a mechanical switch. Relays use a small electric current to turn on a large electric current. For example, a 5-volt signal from a computer could be used to turn on a 240-volt power supply to an electric motor. Relays often serve as a safety mechanism for systems using large currents.

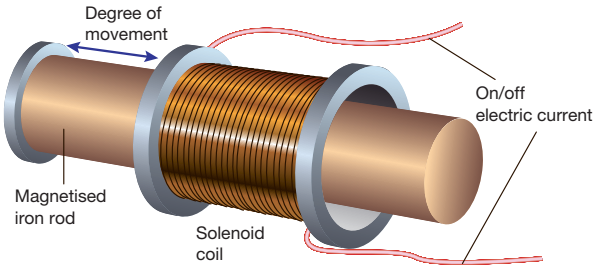


Figure 11.42 A solenoid uses magnetic currents to move a rod.

Pumps

Pumps can be hydraulic (oil) or pneumatic (air). The forces created by a pump make the pistons move, causing a mechanical action. Pumps are often used in robotic arms and can achieve very rapid and precise movements.

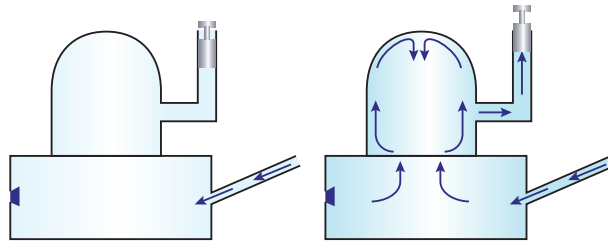


Figure 11.43 A pump pushes gas into a chamber until it is so full that it forces pistons to move.

KNOWLEDGE BUILDER

Understanding pulse wave modulation

Microcontroller boards use **pulse wave modulation (PWM)** to control servos. They need to do this because they produce only digital (on or off) voltage pulses rather than continuous analogue voltage. This works well for regular DC motors, which keep turning while voltage is switched rapidly on and off and also works well with LEDs, which blink too fast to see the flicker, but less well with audio devices. The audio they produce is like the tinny sound of old computer games. The pins of a microcontroller board (such as a micro:bit) only deliver either 3.3V or 0V so a solution is to convert digital to analogue voltage – but how?

How servos use PWM

The solution is a clever one: by switching full voltage on and off very quickly. This technique is called PWM. Hobby servos respond to PWM signals. The length of the signal pulse during one cycle controls the output voltage. Typically, a pulse of 1 ms rotates a servo axle to one extreme, 2 ms to the other extreme, and 1.5 ms sends it to the middle.

In our projects, a pulse is sent every 20 ms and its width is only 1–2 ms. The width of pulse is called its *duty cycle* (expressed as a percentage). This determines the rotational position of the servo shaft as shown in Figure 11.44.

Computational thinking

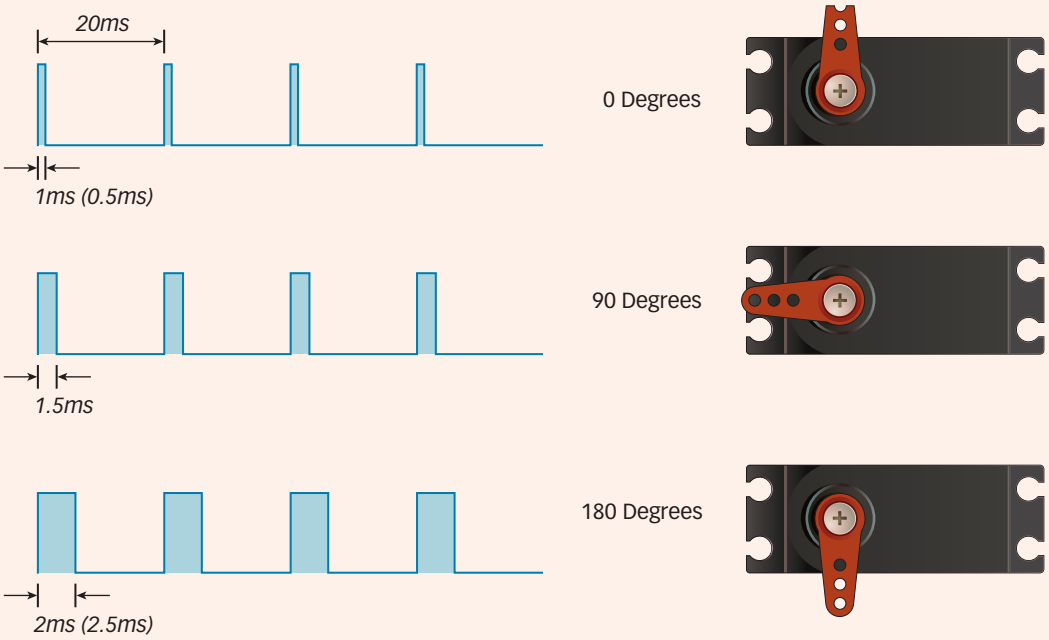


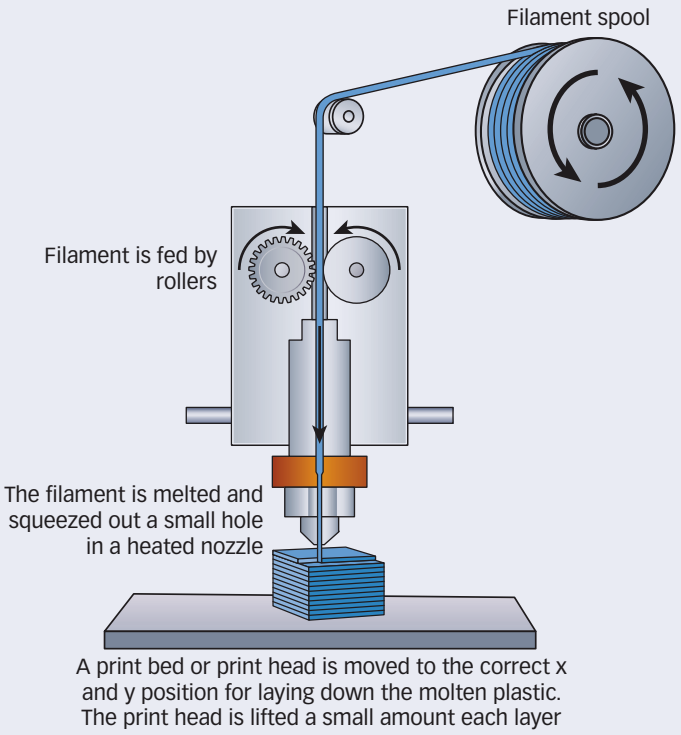
Figure 11.44 Explaining pulse wave modulation (PWM).

CASE STUDY

3D printers and the girl who asked for a frog

Some of the greatest technological mechatronic advances began as toys. In 1988, American Scott Crump's daughter asked him for a toy frog. At first he tried making a frog shape using a glue gun loaded with polyethylene and candle wax by building it up layer by layer. His kitchen ended up with lots of plastic burned onto saucepans. His wife convinced him to move to the garage, where he spent many weekends trying to perfect his methods.





Getty Images/South China Morning Post

Figure 11.45 (a) Inventor, Scott Crump and (b) an illustration of his FDM 3D printer.



Design thinking

He saw his process could be automated by squeezing molten plastic through a tiny nozzle forming objects one layer at a time. In 1989, he patented this idea and started the (now) billion dollar 3D printer company Stratasys. He called his process Fused Deposition Modelling (FDM). FDM melts thermoplastic filament and layers it to create an object directly from files modelled in 3D computer applications.

Scott Crump's success was the result of continuing on when he wasn't certain he would succeed. His belief in his vision provided him with confidence to press on, even though he had only sold five systems over the first three years. He was willing to mortgage his house. He reached a point where he was willing to ask others to help him create a commercial version of his printer and hired a team of plastic, mechanical, computing and electronic engineers.

Ideas that seem silly or trivial can change the world. Steve Jobs, co-founder of Apple computers once said: 'Here's to … the round pegs in the square holes … the ones who see things differently…They change things.' Now the world has 3D printers and it all started with a girl who asked her dad for a frog!

ACTIVITIES: ACTUATORS

1. Identify three common actuators.

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2. Describe the purpose of ONE common actuator.

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3. Distinguish between a servo and a DC motor. Use examples to support your response.

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11.8 Microprocessors

The microprocessor is the heart of a mechatronic system. Its circuits are controlled by a chip which receives input from sensors and sends appropriate signals to actuators. Many of the activities in this chapter use a micro:bit as the microcontroller.

Often the controller of an automatic control system is a computer. Like other systems, it is set up to accept data from input devices, process it and send the data to output devices.

Analogue and digital output pins

Many microcontrollers, including the micro:bit, have connectors that act as either input or output.

On a micro:bit, five large 'pins' (really tabs that connect to 'pins' on the CPU) are labelled 0, 1, 2, 3V and GND. Pins 0, 1 and 2 are known as GPIO, or general-purpose input/output pins, and operate in four different modes depending on how they are coded in the program. These modes are:

Systems thinking

- *Digital output* which delivers either an on ('HIGH') or off ('LOW') (3V or 0V).
- *Digital input* which detects an on or off signal such as from a switch connected to 3V.
- *Analogue output* which uses PWM, a way of using digital pulses to deliver a variable voltage. (PWM was explained in detail in an earlier Knowledge builder.)
- *Analogue input* which is able to measure changing voltages connected to it.

11.9 Control systems

There are two main types of control systems: open loops and closed loop systems. An open-loop control system does not use feedback. An example of an open-loop system is a griller, where the cook chooses the temperature and settings to initiate the cooking cycle. While the food is being cooked, there is no check made by the system on the state of the food. In the manufacturing industry, open-loop automation often involves devices that pick up and place objects.

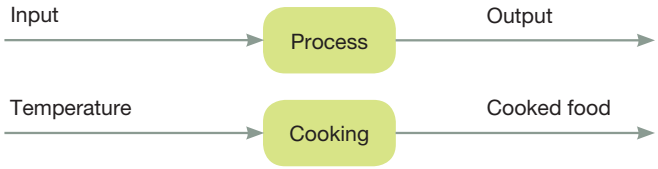


Figure 11.56 An example of how an open-loop system, such as a griller, works.

In contrast, a closed-loop system uses feedback and the output of the system is connected in some way to the input. The data from the output is used to adjust the input. An example is a temperature control system, or thermostat, used to control an air-conditioning unit in a building or to control oven temperature.

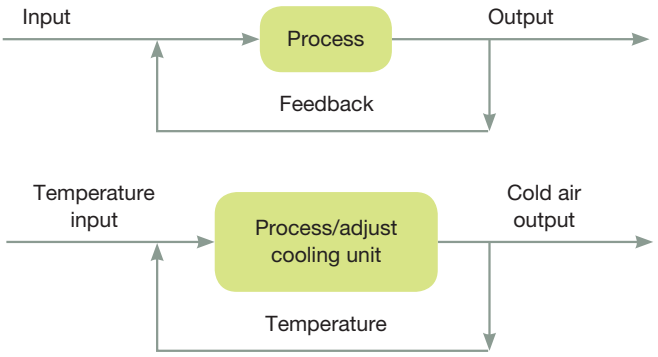


Figure 11.57 An example of a closed-loop system, such as an oven with a thermostat.

In the previous section, you studied two types of motors. A DC motor is an example of an open-loop system. Its input current is not affected by its output rotation. Servos and stepper motors are examples of closed-loop systems. In a servo, the output controls its input as its output shaft is linked through a potentiometer.

11.10 Automated control systems

In our everyday lives, we use devices that are operated by manual control; that is, they need to be operated by humans. Examples include cars, sewing machines, vacuum cleaners and electric drills. Automated systems, on the other hand, are designed to operate under automatic control. 'Automatic control' means they will operate independently or without human intervention. These devices are given appropriate instructions through special programs. Robots can be classified as computer systems that use automated control.

The purpose of an automated control system is to maintain a desired result. It achieves this result by manipulating one or more variables of another system. A good example of this is an air-conditioning unit that uses a thermostat to maintain a constant programmed air temperature. This involves:

- input – the air-conditioned reads the room temperature using a temperature sensor
- process – this reading is compared to the set temperature entered by the user
- output – signals are sent to turn the thermostat on to adjust the temperature.

The working of an automated system

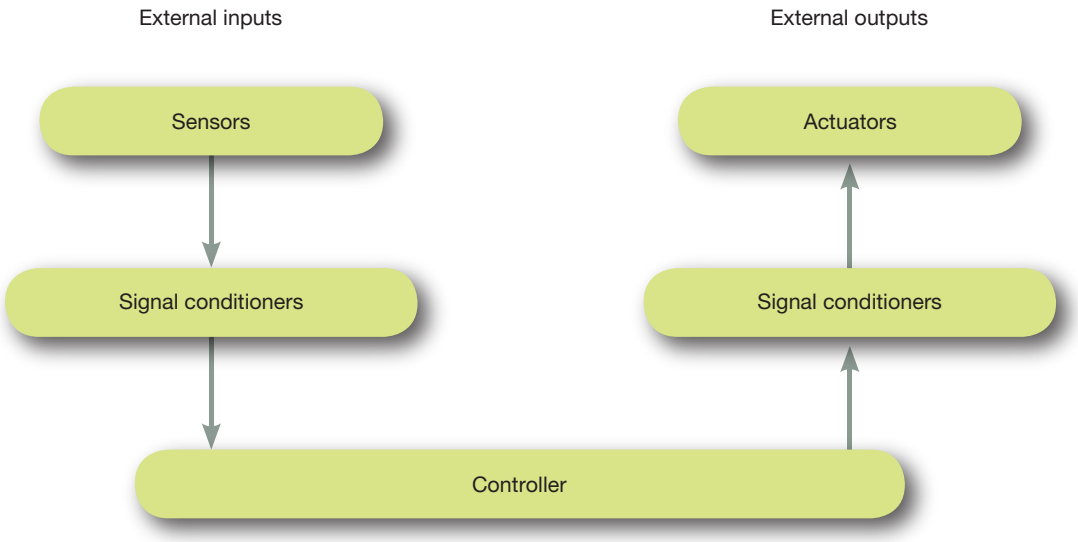


Figure 11.62 A diagrammatic representation of an automated control system.

Table 11.4 The components of an automated control system.

Components	Functions
Sensors	Input devices that detect conditions in the environment then send the results as signals to the controller
Signal conditioners	Devices that help keep the signals passed to and from the controller 'clean'; i.e. reduce the amount of interference in the signals
Controllers	Devices that process the input signals from the sensors to produce the output signals required to drive the actuators
Actuators	Output devices that perform the physical actions that have an effect on the environment

Types of automated control

In the manufacturing industry, a common method of categorising automated control systems is according to the type of operation they perform and the type of products they produce. The three main types of automated control systems are continuous, batch and discrete. The features of each are outlined in Table 11.5.

Table 11.5 Features and examples of different types of control systems.

Control systems	Features
Continuous	Designed and built to perform one manufacturing task, 24 hours a day, 7 days a week, with some time required for system maintenance. Not flexible; for example, a new model of car may require a new assembly plant to be built.
Batch	Produces large quantities of a particular product, but can be quickly altered to produce new or alternative products. The control system is capable of being quickly reprogrammed.
Discrete	Entire system is involved in producing a single item. Produces items only in small quantities. Least suited to large-scale production. Expensive.

Household control

Computer control systems are already a common feature in many households. These include automated:

- security systems
- climate control systems
- integrated home automation systems
- vacuuming
- sprinkler systems
- audio-visual sound systems.

The second project at the end of this chapter offers you the opportunity to create your own automated system.

CASE STUDY

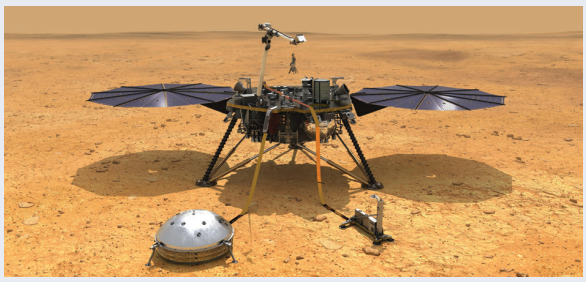
Mars mechatronic 'mole' faces problems

The Insight Mars mission

Before reading on, we suggest you download NASA's free AR application *Spacecraft AR* from your device's app store to your mobile device and select the *InSight* lander under *Mars*. You will be able to explore the landing craft in amazing detail (Figure 11.65), including the subject of this case study: its HP3 hammer probe.

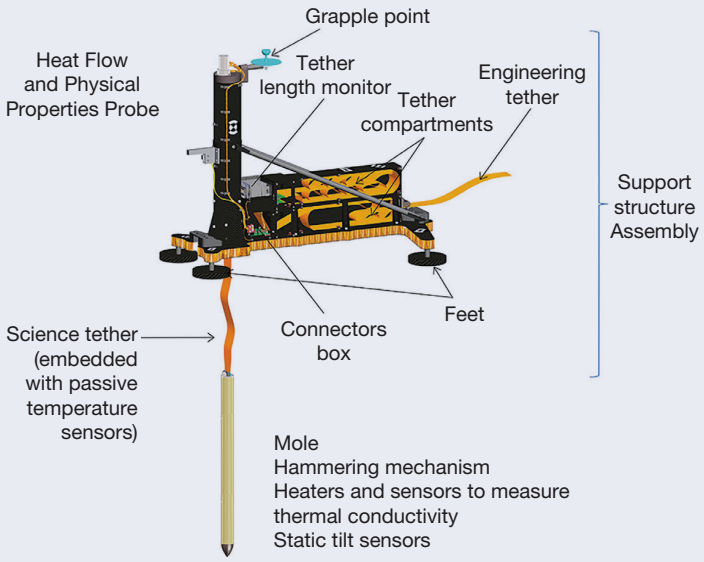
Many earlier Mars missions have scraped soil from the surface of the planet and analysed it, sending the data back to Earth. However, before this InSight Mars mission, no soil has been sampled beneath the surface.

A mechatronic probe, a 'self-hammering nail' (Figure 11.66) called the HP3 and referred to as 'the mole', is designed to burrow five metres below the Martian surface using embedded heat sensors to study thermal properties of Mars' interior, revealing unique information about the planet's geological history.



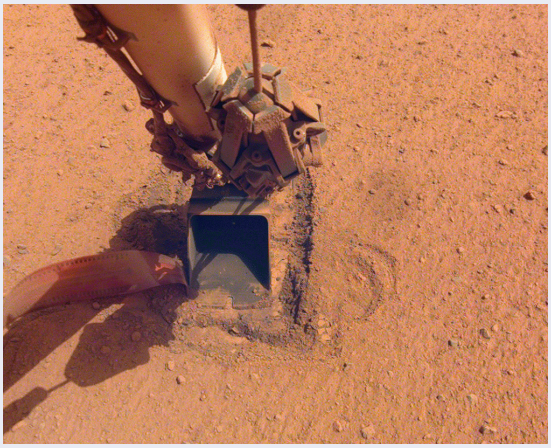
NASA/JPL-Caltech

Figure 11.65 View the Mars InSight landing craft in amazing detail using augmented reality.



NASA/JPL-Caltech/DLR

Figure 11.66 The parts of the HP3 instrument carried to Mars by the InSight lander.



NASA/JPL-Caltech

Figure 11.67 InSight's robotic arm attempting to get the hammer probe digging again by scooping then pressing soil tightly around it.

Other sensors on the device detect echoed vibrations from the hammering, providing data that can be used to determine whether the Mars core is molten or solid.

The data may also reveal if life might have existed on Mars.

Probing problems

After the craft landed, a robotic arm lifted the probe out onto the surface. The principle of operation of the hammer is a cylindrical tube inside which a plunger or hammer is lifted repeatedly then accelerated downwards inside the tube, hitting the tube's pointed end hard, driving it into the soil (see Figure 11.67).

Unfortunately, the probe drilled only 30 centimetres and then refused to go any deeper, tilting at an angle of 15 degrees. Since getting stuck, the mole has hammered 220 times, making slow but steady progress.

Engineers were brought in to problem-solve and over many months they devised a plan.

The plan

They decided they would try to use the robot arm on the InSight lander to first scoop soil around the drill then press against it to create enough friction for it to keep digging. By positioning the edge of the robotic arm on the top rim of the probe's outer shell, they hope to prevent it jumping upwards and hope this will force it to go deeper and eventually engage properly.

The video on the Space.com website explains what is planned.

Design thinking

Weblink
Space.com

Questions

1. Mechatronic engineers often need to solve problems such as this. Can you suggest a design modification for the probe that could help it engage with the surrounding soil?
2. What additional features for the robotic arm might have helped with this problem?

ACTIVITIES: MICROPROCESSORS AND CONTROL SYSTEMS

1. Outline the function of a microprocessor.

2. Distinguish between a closed loop and an open loop control system. Use examples to support your response.

3. Explain how an automatic plant watering system might function if it had a soil moisture sensor and a motorised water pump. Detail input, process and output.

12.1 What is the difference between a robot and an automated system?

There are many overlaps between these terms and in some cases a device could be described as both. Popular culture imagines robots as walking and talking humanoid domestic helpers. This would be a neat way to describe the difference between a **robot** and an **automated system**; however, robots often do not have the ability to move, such as those programmed to weld car parts together. The question is even further confused by some software being referred to as 'bots'!

Another approach is to use the term 'automated system' for systems designed to perform one specific task and the term 'robot' for systems that can be programmed to perform a variety of tasks. In the projects at the end of this chapter, you explore one of each by building a soil irrigation automated system and by programming a robot to escape a maze.

The Skill builders and projects in this chapter provide you with the opportunity to build a wide selection of popular classroom robots and to program them in a variety of computing languages.



Figure 12.2 Robots or mechatronic devices?

12.2 Robotics

Robotics is the science of the use and study of robots. The term 'robot' is usually applied to an automatically guided machine that is able to perform tasks on its own. Robots currently perform many functions, from making cars to defusing bombs. Children and adults play with toy robots, while vacuum-cleaning robots suck up dirt in a growing number of homes.

The use of robots

A robot can be programmed to perform tasks normally carried out by humans. These are often repetitive, dangerous, very precise or even impossible for people to undertake. Robots have the advantage of never getting tired, being able to work with higher accuracy than humans, and even being able to do things humans cannot do, such as working in extreme temperatures. When robots perform boring, repetitive tasks in place of human workers, employees are able to spend more time performing creative, less tedious tasks. However, the disappearance of whole categories of jobs which are more labour intensive or routine will have a significant continuing impact on society.

The history of robots

The earliest robots were built specifically to perform simple tasks on an assembly line. These types of robots are still widely used in manufacturing. Modern robots usually have sensors to accept data from the environment and are increasingly found in the area of artificial intelligence. Intelligent robots are designed to imitate human behaviour and thought processes. Their inputs may include the senses of vision, touch, hearing and smell. An intelligent robot must be able to do two things:

- obtain information from its surroundings
- carry out physical tasks.

Modern robots possess three characteristics:

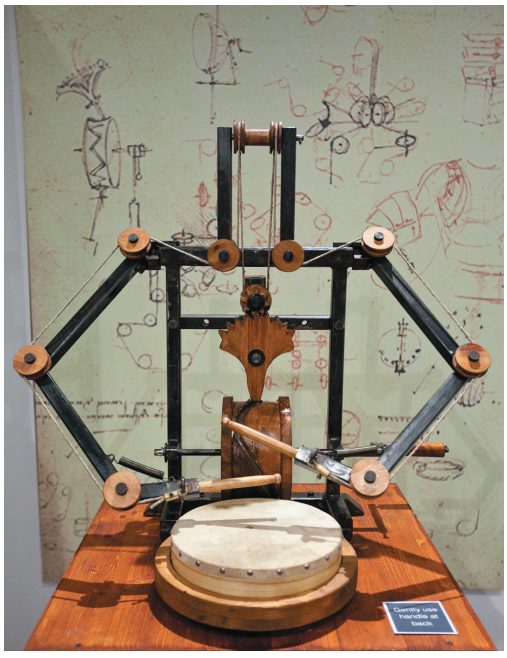
- programmability – they can be programmed to perform tasks like a computer
- mechanical capability – they can perform machine-like tasks in controlled environments
- flexibility – they can perform a variety of tasks by being reprogrammed or by responding to external conditions.

INFOBIT

The word 'robot' comes from the Czech word *robota*, which means 'drudgery' or 'work'.

Mechanical, computerised robots have been around for less than 50 years but the concept of robotic humans is not new. The ancient Greeks spoke of 'mechanical helpers'. In the 15th century, Leonardo da Vinci, the famous Italian mathematician, artist and designer, drew plans for a mechanical drummer (see Figure 12.3). The drummer was for entertainment and it was designed to keep a perfect beat once its handle had been wound. (The project in the previous chapter involves building an automaton like this.)

During the 1920s, Karel Čapek wrote a play called R.U.R. (Rossum's Universal Robots). In Čapek's play, the Rossum factory produced mechanical devices that were called 'robota'. At first, these human-like machines were merely unintelligent slaves, but later, a scientist gave the creatures emotions. The robots eventually turned evil, killing the humans and taking over the world.



Newspix/Charles Brewer

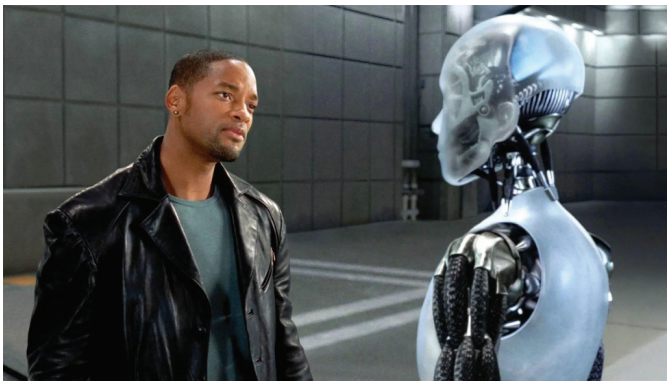
Figure 12.3 Leonardo da Vinci's drumming robot, created by engineer Gabriele Niccolai from da Vinci's sketches.

Following early instances in plays and science-fiction stories, robots started to appear in television shows, such as *Lost in Space*, and have since featured in many movies, including *2001: A Space Odyssey*, *Star Wars*, *Judge Dredd* and *Transformers*.

Asimov's Laws of Robotics

Isaac Asimov (1920–92) took a different view of robots from writers before him. He was the first to give robots relational characteristics in interacting with humans. Asimov thought robots should be regarded as a very important technological innovation. In 1942, he wrote a story about robots, *Runaround*, which contained his three **laws of robotics**. He later added a fourth law and called it the Zeroth Law. These laws are:

- Zeroth Law: A robot may not injure humanity or, through inaction, allow humanity to come to harm.
- First Law: A robot may not injure a human being or, through inaction, allow a human being to come to harm, unless this would violate the Zeroth Law.
- Second Law: A robot must obey orders given to it by human beings, except where such orders would conflict with the Zeroth Law or First Law.
- Third Law: A robot must protect its own existence, as long as such protection does not conflict with the Zeroth Law, First Law or Second Law.



Alamy Stock Photo/Pictorial Press Ltd

Figure 12.4 The ethical issues raised by Asimov's laws of robotics are explored in the movie *I, Robot*, named after an Isaac Asimov collection of short stories.

The robot industry

While the ideas had been around for some time, the construction and use of robots did not properly begin until the mid-1950s.

- In 1956, the first robot company was formed by George Devol and Joseph Engelberger.
- In 1959, computer-assisted manufacturing was first demonstrated at the Massachusetts Institute of Technology.
- In 1961, the first industrial robot, called UNIMATE, came online in a General Motors car factory.



Alamy Stock Photo/kolvenbach

Figure 12.5 In 1999, Sony built its first AIBO (robotic dog). The final version, released in 2006, could speak 1000 words, react to an owner's commands, take photographs and play music.

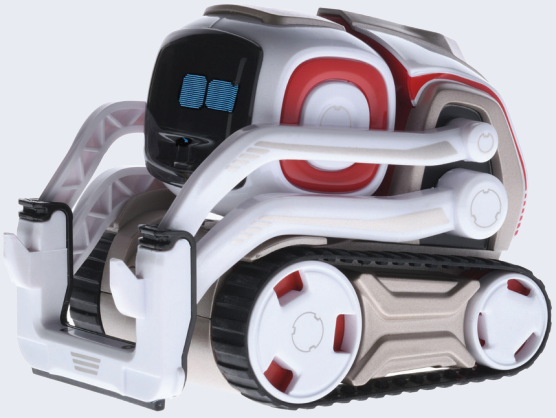


Alamy Stock Photo/Broccoli Photography

Figure 12.6 When robots first appeared in movies, they were often depicted as unfeeling villains with the potential to run amok and take over the world. More recently, robots have been portrayed as intelligent, social beings with a range of human-like emotions, as in the case of Pixar's robotic hero Wall-e.

CASE STUDY

An AI buddy



Alamy Stock Photo/Hugh Threlfall



Shutterstock.com/JCDH

Figure 12.7 (left) Anki's Cozmo, a programmable robot with built-in machine learning; (right) the Roomba robotic vacuum cleaner.

A companion robot everyone wants?

Anki, maker of Vector, a toy-like autonomous countertop robot, is shutting down, and hundreds of people are losing their jobs ... Sure, home robots like Vector are cute, but is this something consumers actually want? Apparently not enough customers for Anki, but here's what's far less clear: Beyond the success of Roomba, what on earth do we want sophisticated machines to do for us around the house?

There's perhaps no more challenging place to do that than in the pure chaos of the home, what roboticists call an "unstructured" environment. Robots have for decades ruled the factory, because that's a "structured" environment that's easy to master. They can travel along prescripted paths and perform highly repetitive maneuvers while facing few, if any, unexpected obstacles. In the home, robots have to contend with stairs and children and clutter on the floor.

The popularity of assistants like Amazon's Alexa didn't make the company's job any easier. Like the Roomba and its sweeping, these assistants do one thing very well. And they do it by sitting stationary on a countertop, which means they don't have to navigate the chaos of the home. If home robots can't yet grab you a beer from the fridge, fold laundry, or pick up literally any object, is there much value in their mobility? And if they're not as smart as Alexa, is there much point in making friends with them?

The makers of home robots also struggle to keep up with Amazon's and Google's ability to iterate on their voice assistants quickly. Hardware like Vector's is an entirely different story. If the company designs a robot and pushes it out to market and it's not working as desired, it could take years to redesign it.

The ultimate goal for home robots may not even be to build you a new best buddy but to make robots in a sense disappear into the home. "In the case of the Roomba, it's just a vacuum cleaner, we don't think about it much as a robot anymore," says Cory Kidd, developer of Mabu, a medical robot that interacts with patients. "And I think that's actually a success, when people think of these things in terms of their functionality and the problem they're trying to solve." Think about the microprocessor: If you wear an old digital watch like me, you don't think of it as a computer because it's become more a part of you than something unique. But it's indeed a computer.

Still, what home robots do have going for them is that novel charm. This is where the human-robot interaction bit gets really interesting. Watching Kuri and Vector roll around beeping and booping and following your commands was a fascinating thing to behold. In essence, their physical presence allows for a bond with a machine loaded with far more charisma than what a stationary voice assistant can muster. That's the promise of home robots: The illusion of agency may not make them particularly useful at the moment, but they're forging a strange new relationship between people and machines. No longer are robots just tools—they're friends. And maybe one day (perhaps once they've sprouted arms and hands) they'll be so ingrained in the home, like Roomba, that we'll forget they're robots at all.

Vector may have failed to win enough hearts to survive, but Anki has made invaluable progress in fleshing out what this new human-machine relationship might look like. "They were on the forefront of social robot design, and poured so much into getting the interaction right," says MIT roboticist Kate Darling, who studies human-robot interaction. "It was a beautiful product. But they had trouble growing it from nifty toy to something else."

ACTIVITIES

Identify

1 Define the term 'robotics'.
2 Identify the purpose of the earliest robots.
3 Identify the category of component in modern robots that enables them to accept data from the environment.
4 Name two characteristics of an intelligent robot.
5 Identify three main features of modern robots.
6 What was the name of the first industrial robot and in what year did it come online?

Analyse

7 Scientists have a very different view of robots from science-fiction writers and movie-makers. Describe these differences.
8 Discuss the relevance of Asimov's Laws of Robotics to the robotics industry today.
9 Describe a way to distinguish robots from automated systems.

Extend

10 Either write a short story involving Asimov's Laws of Robotics or summarise the plot of *I, Robot*, the movie starring Will Smith, loosely based on an Asimov story.

12.3 Types of robots

There are many different types of robots. One simple way to classify robots is according to whether they are stationary or mobile. Stationary robots are fixed in place, but their arms can reach out and manipulate parts and tools. These robotic arms have an **end effector** (Figure 12.8) attached to carry out a task.

Stationary robots are often found in assembly lines, where they perform simple tasks such as placing objects on a conveyor belt. Mobile robots, on the other hand, can move around in their environment and are not fixed to one physical location.

Industrial robots

Industrial robots have been around for more than 50 years, performing simple, repetitive tasks with superhuman speed and precision. They are now responsible for a remarkably wide variety of tasks, from assembling computers to popping frozen dinners into trays.

In the manufacturing industry, robots perform a variety of tasks (Figure 12.9), including:

- welding
- spray-painting

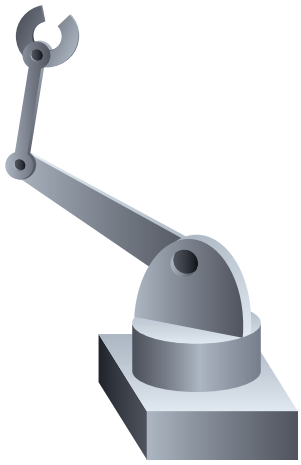


Figure 12.8 A robotic arm with an end effector attached.

- assembly operations
- palletising and material handling
- dispensing operations
- laboratory applications.

Robots with exchangeable tools

Some robots used in manufacturing are designed to perform one particular task, while others are designed with interchangeable parts, which make them more versatile. For example, a robot may act as a welder for one task, but by changing the end gripper, it may be turned into a spray painter. To reprogram a new robot arm, an operator simply selects from among choices that the robot itself offers; a worker can teach a robot's 'eyes' to recognise a new part in less than five minutes.



Thinkstock/Andrzej Thiel

Figure 12.9 Robots in manufacturing.

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Industrial accidents and the prevalence of some cancers have declined as a result of robots performing hazardous jobs in the workplace.

Mobile robots

Mobile robots usually move around on wheels, tracks or legs and began to appear in the 1980s in the area of nuclear science. In 1979, a nuclear accident at Three Mile Island in the United States caused a leak of radioactive material. The accident was a major concern because of the amount of radioactive substances released into the environment. This incident led to the development and use of special robots, called 'teleoperators', to handle the radioactive material. Today, mobile robots are used in areas such as security control, bomb disposal, planet exploration, transporting hazardous materials, and disaster rescue and recovery (Figure 12.10).



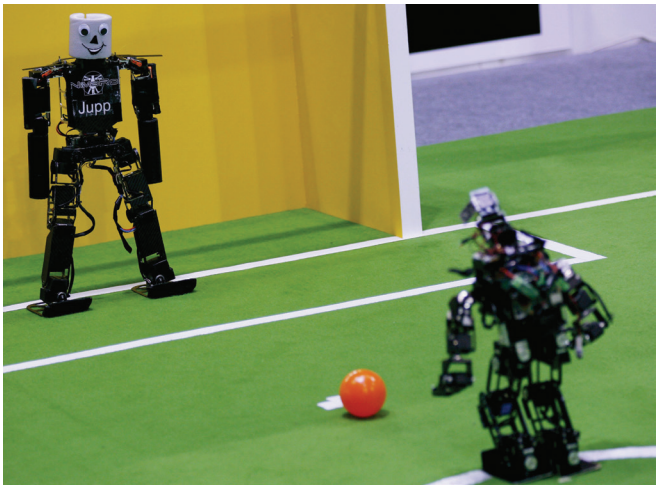
Hydronalix

Figure 12.10 This mobile robot is used to rescue distressed swimmers. Its sonar device scans for the underwater movements of swimmers in trouble and an in-built speaker allows a lifeguard on shore to calm and instruct the person being rescued.

Educational robots

Robots, or robotic kits, are used extensively in education. Robotic equipment in the classroom is used to teach and learn about programming and computer control. A well-known example are the EV3 and Spike Prime robot systems developed by LEGO. These comprise a brick control device, sensors and motors, which users assemble into models. Among other functions, the robots can be programmed to walk and to carry out a variety of tasks or missions. In addition, projects in this chapter will provide you and your classmates with the opportunity to collaboratively embark on a series of robotic challenges.

Another educational example is the programming of robots to play soccer in the RoboCup competition (Figure 12.11). Teams from all over the world enter the competition. The robots operate completely autonomously. The rules stipulate there must be no external control by humans.



Getty Images/Malte Christians

Figure 12.11 In the RoboCup Humanoid League, autonomous robots with human-like senses play soccer against each other. RoboCup is an annual international robotics competition that promotes robotics and artificial intelligence (AI) research. There are many other stages of the competition, such as RoboCup Rescue, RoboCup OnStage (Performance) and RoboCup Junior.



Source: Wired

Figure 12.13 Many modern toys are considered robotic because they are programmed devices. They may react to light, sound and touch, using special sensors for input.

INFOBIT

Interactive 'social robots' called 'Petimos' have been designed to help protect children when they make friends on social networks. Petimos' primary function is to guard against cyber bullies and adults masquerading as children.

Domestic robots

Robotic toys include remote-control cars, remote-control boats, barking dogs and crying dolls (Figure 12.13). Other domestic robots perform boring and repetitive household chores, such as vacuuming (Figure 12.14), or provide services such as security surveillance (Figure 12.12).

In Japan, domestic robots have been developed which provide companionship and assistance to the elderly.



Shutterstock.com/lev radin

Figure 12.12 The K5 security robot developed by US firm, Knightscope.



Source: Kompai

Figure 12.14 Kompai is a domestic robot that can record grocery lists via voice commands, call an emergency number for medical assistance and remind elderly people to take their medication.

CASE STUDY

The changing nature of work

Spill something in a nuclear reactor core and you can't just send in the janitor to clean it up. (Well you can, but just once!) What you can do – repeatedly – is send in remotely operated robots designed by Matt Cole. At 33, Cole is chief engineer of Colorado-based SA Technology, which earns more than $20 million a year making 'bots' for nuclear facilities.

For a kid who taught himself to weld, mill and lathe before he was old enough to vote, Cole believes it is a dream gig. 'We do the worst, the hottest, the weirdest jobs,' he says.

Like what? For starters, his creations mop up areas that are so radioactive they've been dubbed infinity rooms – a Geiger counter needle goes past 11!

Each task requires its own unique bot. For example, if a piece of nickel lodges in a pipe at a reactor plant and turns into the extremely radioactive and dangerous isotope cobalt-60, Cole and his team of engineers construct an agile bot that can extend a telescopic arm, spray liquid nitrogen to cool the pipe, and cut away the radioactive segment. Other bots are designed to plunge into giant waste tanks and hose down radioactive materials such as caesium-137.

SA Technology bots are at every major nuclear weapons facility in the United States, as well as most nuclear power plants and as more and more Cold War-era weapons facilities are being decommissioned, business has never been better.

Joshua Zaffos, 'Alphageek: Matt Cole Builds Robo Rooters that Scour Out Reactors', Wired magazine, June 2010



Wired/Joshua Zaffos

Figure 12.15 Matt Cole's robots perform tasks that are too dangerous for humans.

INFOBIT

During the 2011 nuclear crisis in Japan, many people were surprised that robots were not deployed to repair the crippled reactors. Despite being a technological powerhouse and a world leader in robotics, Japan essentially relied on a small team of engineers, firefighters and helicopter pilots to perform the task of cooling the reactors.



Alamy Stock Photo/ZUMA Press, Inc.

Figure 12.16 In order to reduce the safety risks to humans, robots can venture into radiation-exposed areas.

Questions

1. Describe the tasks carried out by Matt Cole's robots.
2. Are robots such as those created by Matt Cole the answer for solving toxic accidents and decommissioning nuclear weapons?
3. Automation and the development of robots for dangerous tasks are viewed by most people as beneficial to society. However, many workers in less dangerous jobs have been made redundant by robots. What is your response to this issue?



Figure 12.17 The *Tobbie* robot won an Educational Toy of the Year award. It has an IR sensor for input, an audio buzzer and three motors as actuators, controlled by a micro:bit microcontroller board allowing it to be programmed using either block coding or Python.

12.8 Ethical issues in robotics

Machines outsmarting humans?

Consider these creations:

- a robot that can open doors and find electrical outlets to recharge itself
- computer viruses no one can stop
- predator drones that can drop bombs autonomously.

Impressed and alarmed by advances in artificial intelligence, a group of computer scientists is debating whether there should be limits on research that might lead to loss of human control over automated systems that carry a growing share of society's workload, from waging war to chatting with customers on the phone.

Their concern is that further advances could create profound social disruptions and even have dangerous consequences. As examples, the scientists pointed to a number of technologies as diverse as experimental medical systems that interact with patients to simulate empathy and computer viruses that defy extermination and could thus be said to have reached a 'cockroach' stage of machine intelligence.

While the computer scientists agreed that we are a long way from HAL (the computer that took over the spaceship in Stanley Kubrick's famous movie *2001: A Space Odyssey*), they said there was real concern that technological progress would transform the workforce by destroying a widening range of jobs, as well as force people to learn to live with machines that increasingly copy human behaviours.

The researchers dismissed the possibility of highly centralised super intelligences and the idea that intelligence might spring spontaneously from the Internet. But they agreed that robots that can kill autonomously raise important questions about military personnel in a distant control room being sheltered from the consequences of their actions.

There are other concerns too. What could a criminal do with a speech synthesis system that could mimic humans? What might be the implications if AI was used to mine personal information from smartphones?



Science Photo Library/US AIR FORCE

Figure 12.46 Predator drones, such as this one photographed in Afghanistan, still need a human hand to work – at least for now.

Chapter review quiz

CASE STUDY

OpenAI

OpenAI has trained AI language models using billions of pieces of data to such a level that it is now good at understanding and generating text when given a prompt. The prompt 'Suggest three names for a koala that is a superhero' delivered '*Super Koala, Captain Cuddles, The Mighty Marsupial*'.

OpenAI can even generate accurate lines of computer code in Python and other languages when given a simple task.

ChatGPT

ChatGPT is AI driven software powered by *OpenAI* that uses machine learning to respond to questions asked by users.



Shutterstock.com/FAMILY STOCK

Figure 12.47 *ChatGPT* can generate text which is convincingly human.

It can generate impressive responses to any fact-based query. Educators have expressed concern that complete essays on any topic can be passed off as the work of students, compromising any essay work required to be done at home. Many school systems around the world have blocked its use within the school.

Robot engineers are working on robots that will integrate such technology into their robots.

Dall·E

DALL·E 2 is an *OpenAI* powered system that can create realistic images and art from any description provided in natural language. You can try this out yourself using its online website. Can you imagine ways this type of technology might be used in robotics?

ROBOTICS

1. Discuss the relevance of Asimov's laws of robotics to the robotics industry today.

2. Discuss the impact of technological advancement on the workforce. Use relevant examples to support your response.

GLOSSARY OF TERMS

Accelerometer

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Actuator

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Automated Control System

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Closed-Loop Control System

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Control System

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Data Logger

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Data Logging

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End Effector

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Isaac Asimov's Laws of Robotics

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Mechatronics

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Microcontroller

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Microprocessor

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Open-Loop Control System

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Robotics

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Sensor

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Servo

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